信 息 工 程 学 院

《云计算》实验报告

学院：信息工程学院　**班级**：数科 2102　**姓名**：陈　**学号**:202101　**成绩**:

一、实验题目

HDFS 编程

二、实验目的

通过本实验掌握 HDFS 基本操作；掌握利用 Hadoop 提供的 Java API 进行编程对 HDFS 进行操作。

三、实验平台

希冀平台

四、实验过程及步骤

实验 2.1

使用 ssh 命令访问大数据集群，查看集群配置

实验报告的内容与格式按任课教师的要求书写。

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

-bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
root@master:~# exit
logout
Connection to master closed.
root@cg:~/Desktop# ssh slave1
Warning: Permanently added the ECDSA host key for IP address '100.64.0.41' to th
e list of known hosts.
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 5.4.0-150-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

-bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
root@slave1:~# exit
logout
Connection to slave1 closed.
root@cg:~/Desktop# cat /etc/hosts
127.0.0.1       localhost
::1     localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
100.64.0.39 cg
100.64.0.40 master
100.64.0.41 slave1
100.64.0.42 slave2
100.64.0.43 slave3
root@cg:~/Desktop#
```

启动 hadoop 服务

```
root@master:~# bash /scripts/hadoop/start-hadoop.sh
ZooKeeper JMX enabled by default
Using config: /opt/module/zookeeper-3.4.10/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED
Warning: Permanently added the ECDSA host key for IP address '100.64.0.41' to the list of known hosts.
ZooKeeper JMX enabled by default
Using config: /opt/module/zookeeper-3.4.10/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED
Warning: Permanently added the ECDSA host key for IP address '100.64.0.42' to the list of known hosts.
ZooKeeper JMX enabled by default
Using config: /opt/module/zookeeper-3.4.10/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED
Warning: Permanently added the ECDSA host key for IP address '100.64.0.43' to the list of known hosts.
ZooKeeper JMX enabled by default
Using config: /opt/module/zookeeper-3.4.10/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED
Stopping namenodes on [master]
master: Warning: Permanently added the ECDSA host key for IP address '100.64.0.40' to the list of known hosts.
master: no namenode to stop
slave2: no datanode to stop
slave3: no datanode to stop
slave1: no datanode to stop
Stopping secondary namenodes [0.0.0.0]
0.0.0.0: no secondarynamenode to stop
Starting namenodes on [master]
```

验证

```
root@master:~# jps
3330 Jps
3234 JobHistoryServer
1251 QuorumPeerMain
2875 ResourceManager
2443 SecondaryNameNode
2111 NameNode
```

```
root@slave1:~# jps
912 Jps
337 DataNode
535 NodeManager
173 QuorumPeerMain
root@slave1:~#
```

创建工作目录

```
root@cg:~/Desktop# mkdir -p ~/Desktop/workspace/hdfs_op
root@cg:~/Desktop# cd ~/Desktop/workspace/hdfs_op
root@cg:~/Desktop/workspace/hdfs_op#
```

查看目录

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -ls /
Found 1 items
drwxrwx---   - root supergroup          0 2023-10-11 08:13 /tmp
```

创建目录

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -mkdir /newDir
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -ls /
Found 2 items
drwxr-xr-x   - root supergroup          0 2023-10-11 08:18 /newDir
drwxrwx---   - root supergroup          0 2023-10-11 08:13 /tmp
root@cg:~/Desktop/workspace/hdfs_op#
```

删除目录

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -rm -r /newDir
23/10/11 08:19:03 INFO fs.TrashPolicyDefault: Namenode trash configuration: Deletion interval = 0 minutes, Emptier inter
val = 0 minutes.
Deleted /newDir
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -ls /
Found 1 items
drwxrwx---   - root supergroup          0 2023-10-11 08:13 /tmp
root@cg:~/Desktop/workspace/hdfs_op#
```

文件上传

```
root@cg:~/Desktop/workspace/hdfs_op#
root@cg:~/Desktop/workspace/hdfs_op# pwd
/headless/Desktop/workspace/hdfs_op
root@cg:~/Desktop/workspace/hdfs_op# echo "hello" > hello.txt
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -mkdir /newDir
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -put hello.txt /newDir
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -ls /newDir
Found 1 items
-rw-r--r--   4 root supergroup          6 2023-10-11 08:23 /newDir/hello.txt
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -cat /newDir/hello.txt
hello
root@cg:~/Desktop/workspace/hdfs_op#
```

文件下载

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -get /newDir/hello.txt hello_get.txt
23/10/11 08:24:54 WARN hdfs.DFSClient: DFSInputStream has been closed already
root@cg:~/Desktop/workspace/hdfs_op# cat hello_get.txt
hello
root@cg:~/Desktop/workspace/hdfs_op#
```

查看文件内容

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -cat /newDir/hello.txt
hello
root@cg:~/Desktop/workspace/hdfs_op#
```

查看文件大小

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -du /newDir
6  /newDir/hello.txt
root@cg:~/Desktop/workspace/hdfs_op# ls -l hello.txt
-rw-r--r-- 1 root root 6 Oct 11 08:22 hello.txt
root@cg:~/Desktop/workspace/hdfs_op#
```

复制文件

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -cp /newDir/hello.txt /newDir/hello_cp.txt
23/10/11 08:28:00 WARN hdfs.DFSClient: DFSInputStream has been closed already
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -cat /newDir/hello_cp.txt
hello
root@cg:~/Desktop/workspace/hdfs_op#
```

文件重命名

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -mv /newDir/hello.txt /newDir/hello_mv.txt
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -ls /newDir
Found 2 items
-rw-r--r--   4 root supergroup          6 2023-10-11 08:28 /newDir/hello_cp.txt
-rw-r--r--   4 root supergroup          6 2023-10-11 08:23 /newDir/hello_mv.txt
```

文件权限修改

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -ls /newDir
Found 2 items
-rw-r--r--   4 root supergroup          6 2023-10-11 08:28 /newDir/hello_cp.txt
-rw-r--r--   4 root supergroup          6 2023-10-11 08:23 /newDir/hello_mv.txt
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -chmod 777 /newDir/hello_mv.txt
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -ls /newDir
Found 2 items
-rw-r--r--   4 root supergroup          6 2023-10-11 08:28 /newDir/hello_cp.txt
-rwxrwxrwx   4 root supergroup          6 2023-10-11 08:23 /newDir/hello_mv.txt
```

删除文件

```
-rw-r--r--   4 root supergroup          6 2023-10-11 08:28 /newDir/hello_cp.txt
-rwxrwxrwx   4 root supergroup          6 2023-10-11 08:23 /newDir/hello_mv.txt
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -rm /newDir/hello_mv.txt
23/10/11 08:30:53 INFO fs.TrashPolicyDefault: Namenode trash configuration: Deletion interval = 0 minutes, Emptier inte
val = 0 minutes.
Deleted /newDir/hello_mv.txt
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -ls /newDir
Found 1 items
-rw-r--r--   4 root supergroup          6 2023-10-11 08:28 /newDir/hello_cp.txt
root@cg:~/Desktop/workspace/hdfs_op#
```

查看文件系统信息

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -df -h
Filesystem             Size    Used  Available  Use%
hdfs://master:9000  21.7 T  84.0 K     13.7 T    0%
root@cg:~/Desktop/workspace/hdfs_op#
```

作业

1、 使用 hdfs 命令在 hdfs 文件系统上创建文件夹`/mydir`；

```
root@cg:~/Desktop/workspace/hdfs_op# hadoop fs -mkdir /mydir
root@cg:~/Desktop/workspace/hdfs_op#
```

2、 在 master 节点上以自己的学号作为文件名创建一个文件，文件的内容是一小段随意的英文；

```
root@master:~# echo "wo yao wan yuan shen" > 2021013299.txt
```

3、 使用 hdfs 命令将该文件上传到 hdfs 文件系统上的`/mydir`文件夹下；

```
root@master:~# hadoop fs -put 2021013299.txt /mydir
root@master:~# hadoop fs -ls /mydir
Found 1 items
-rw-r--r--   4 root supergroup         21 2023-10-11 08:36 /mydir/2021013299.txt
root@master:~#
```

4、 使用 hdfs 命令浏览该文件的内容；

```
root@master:~# hadoop fs -cat /mydir/2021013299.txt
wo yao wan yuan shen
root@master:~#
```

5、 将该文件重命名为 `homework.txt`；

```
root@master:~# hadoop fs -mv /mydir/2021013299.txt /mydir/homework.txt
root@master:~# hadoop fs -ls /mydir
Found 1 items
-rw-r--r--   4 root supergroup         21 2023-10-11 08:36 /mydir/homework.txt
root@master:~#
```

6、 将该文件的权限修改为仅自己可以读写，即 `rw-------`；

```
root@master:~# hadoop fs -ls /mydir
Found 1 items
-rw-r--r--   4 root supergroup         21 2023-10-11 08:36 /mydir/homework.txt
root@master:~# hadoop fs -chmod 600 /mydir/homework.txt
root@master:~# hadoop fs -ls /mydir
Found 1 items
-rw-------   4 root supergroup         21 2023-10-11 08:36 /mydir/homework.txt
root@master:~#
```

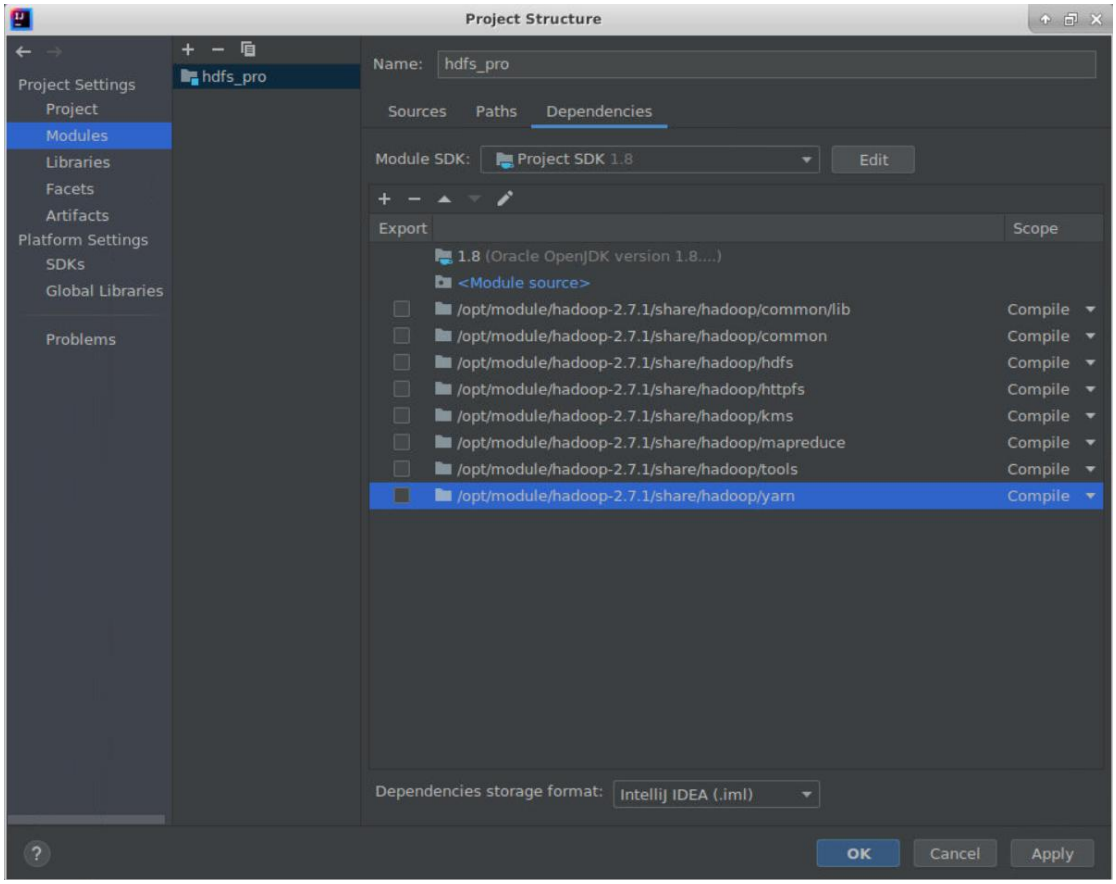
7、将上传的文件删除。

```
root@master:~# hadoop fs -rm /mydir/homework.txt
23/10/11 08:42:56 INFO fs.TrashPolicyDefault: Namenode trash configuration: Deletion interval = 0 minutes, Emptier inter
val = 0 minutes.
Deleted /mydir/homework.txt
^[[Aroot@master:~# hadoop fs -ls /mydir
root@master:~#
```

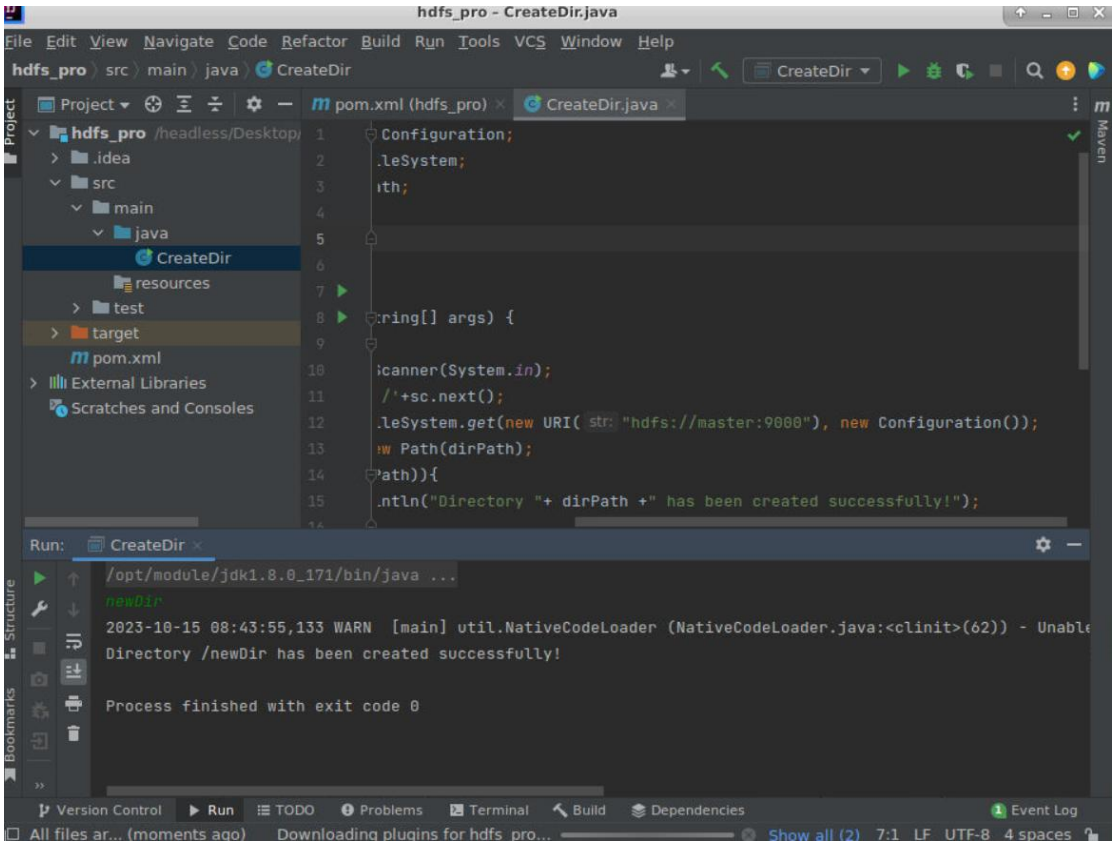
实验 2.2

环境配置

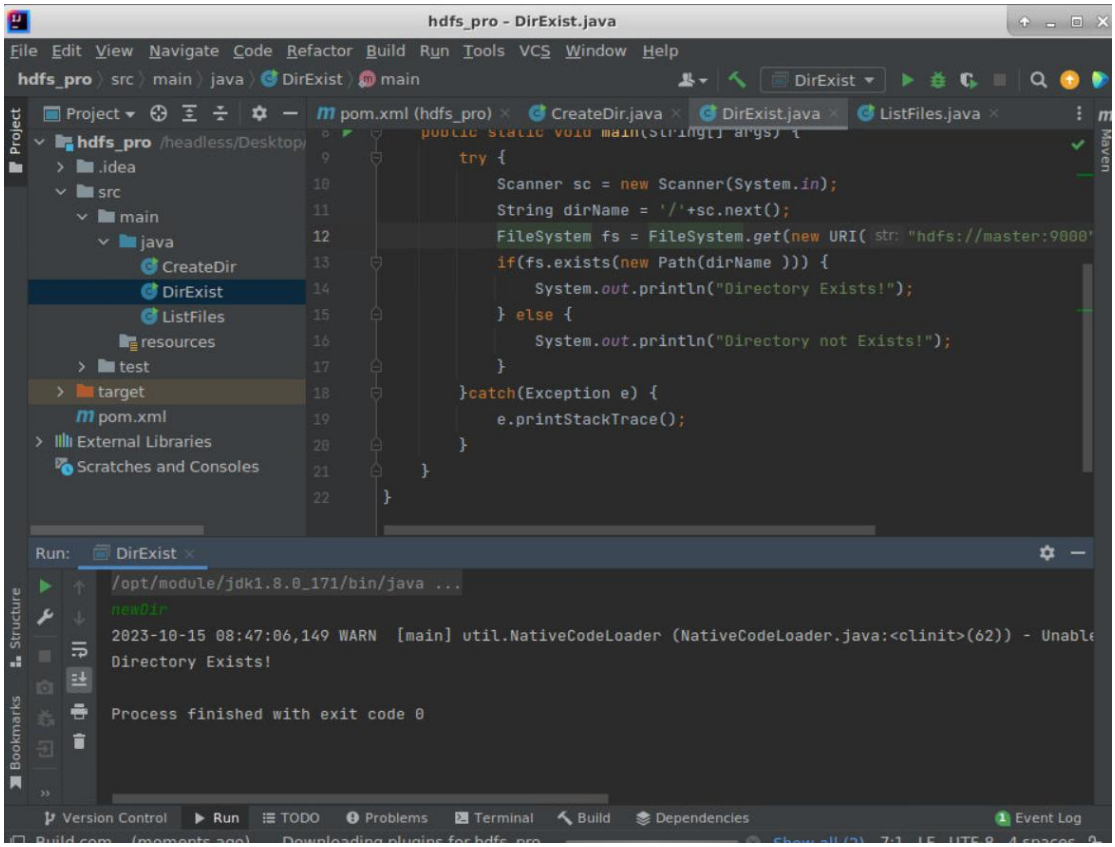
```
starting yarn daemons
starting resourcemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root-res
ourcemanager-master.out
slave1: starting nodemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root
-nodemanager-slave1.out
slave2: starting nodemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root
-nodemanager-slave2.out
slave3: starting nodemanager, logging to /opt/module/hadoop-2.7.1/logs/yarn-root
-nodemanager-slave3.out
no historyserver to stop
starting historyserver, logging to /opt/module/hadoop-2.7.1/logs/mapred-root-his
toryserver-master.out
root@master:~# jps
1281 SecondaryNameNode
84 QuorumPeerMain
948 NameNode
2184 Jps
2088 JobHistoryServer
1721 ResourceManager
root@master:~#
```

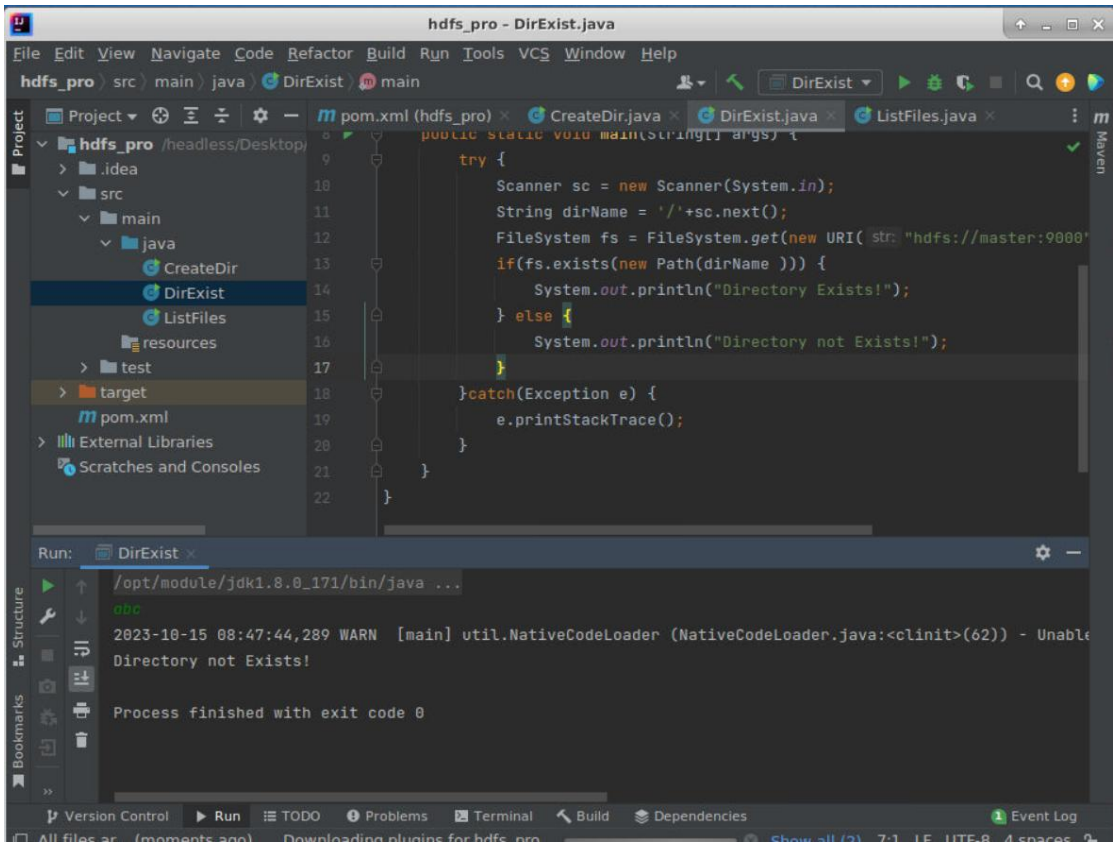

Project Structure
Project Settings
Project
Modules
Libraries
Facets
Artifacts
Platform Settings
SDKs
Global Libraries
Problems
hdfs_pro
Name: hdfs_pro
Sources
Paths
Dependencies
Module SDK: Project SDK 1.8
Edit
Export
Scope
1.8 (Oracle OpenJDK version 1.8....)
<Module source>
/opt/module/hadoop-2.7.1/share/hadoop/common/lib Compile
/opt/module/hadoop-2.7.1/share/hadoop/common Compile
/opt/module/hadoop-2.7.1/share/hadoop/hdfs Compile
/opt/module/hadoop-2.7.1/share/hadoop/httpfs Compile
/opt/module/hadoop-2.7.1/share/hadoop/kms Compile
/opt/module/hadoop-2.7.1/share/hadoop/mapreduce Compile
/opt/module/hadoop-2.7.1/share/hadoop/tools Compile
/opt/module/hadoop-2.7.1/share/hadoop/yarn Compile
Dependencies storage format: IntelliJ IDEA (.iml)
OK
Cancel
Apply

建立文件夹

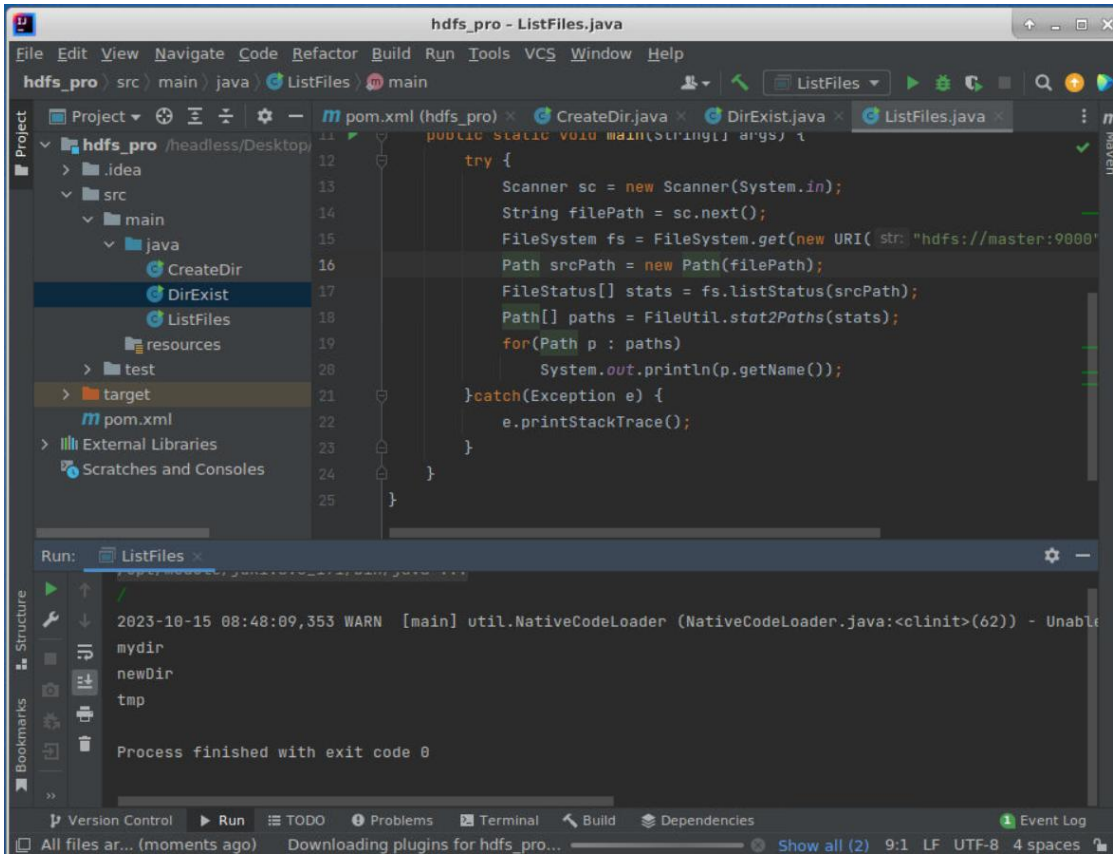

hdfs_pro - CreateDir.java
File Edit View Navigate Code Refactor Build Run Tools VCS Window Help
hdfs_pro src main java CreateDir
pom.xml (hdfs_pro)
CreateDir.java
hdfs_pro /headless/Desktop/
.idea
src
main
java
CreateDir
resources
test
target
pom.xml
External Libraries
Scratches and Consoles
Configuration;
.leSystem;
ith;
:ring[] args) {
icanner(System.in);
/'+sc.next();
.leSystem.get(new URI(str: "hdfs://master:9000"), new Configuration());
w Path(dirPath);
ath)){
.ntln("Directory "+ dirPath +" has been created successfully!");
Run: CreateDir
/opt/module/jdk1.8.0_171/bin/java ...
newDir
2023-10-15 08:43:55,133 WARN [main] util.NativeCodeLoader (NativeCodeLoader.java:<clinit>(62)) - Unable
Directory /newDir has been created successfully!
Process finished with exit code 0
Version Control
Run
TODO
Problems
Terminal
Build
Dependencies
Event Log
All files ar... (moments ago)
Downloading plugins for hdfs_pro...
Show all (2)
7:1
LF
UTF-8
4 spaces

检测文件夹是否存在

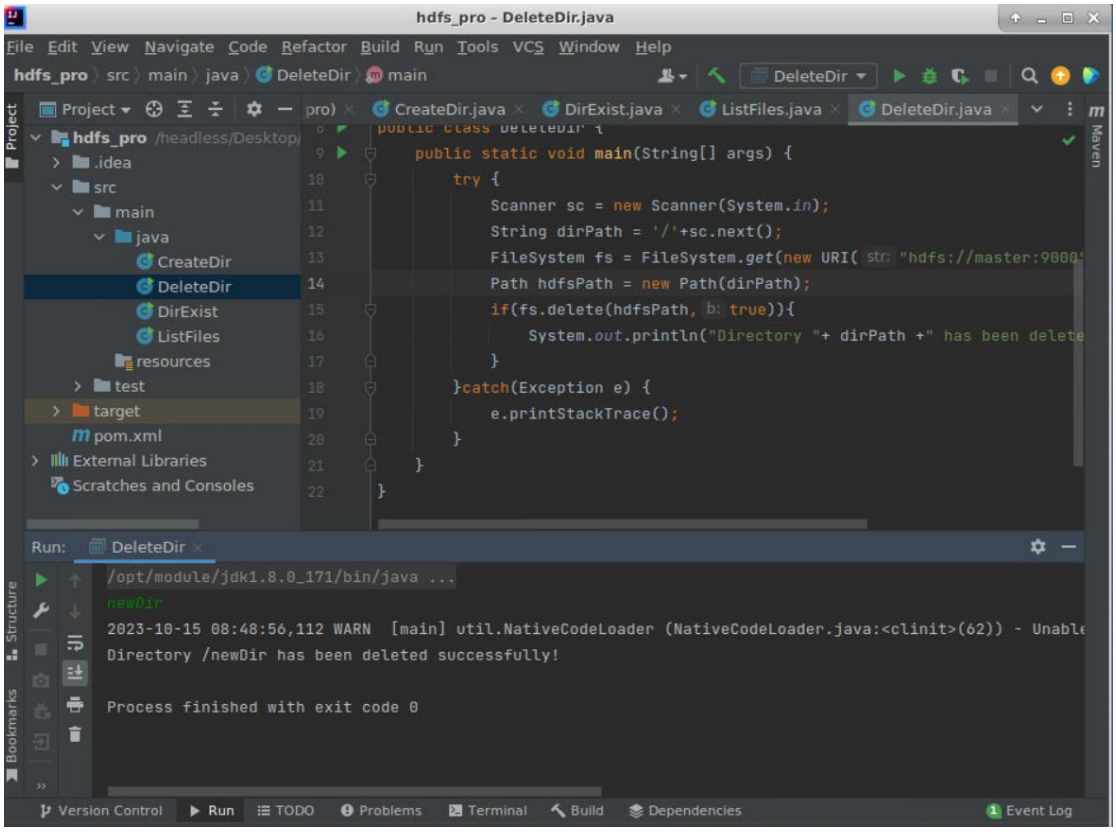

hdfs_pro - DirExist.java
File Edit View Navigate Code Refactor Build Run Tools VCS Window Help
hdfs_pro src main java DirExist main
Project
hdfs_pro /headless/Desktop/
.idea
src
main
java
CreateDir
DirExist
ListFiles
resources
test
target
pom.xml
External Libraries
Scratches and Consoles
pom.xml (hdfs_pro)
CreateDir.java
DirExist.java
ListFiles.java
try {
Scanner sc = new Scanner(System.in);
String dirName = '/'+sc.next();
FileSystem fs = FileSystem.get(new URI(str: "hdfs://master:9000"
if(fs.exists(new Path(dirName))) {
System.out.println("Directory Exists!");
} else {
System.out.println("Directory not Exists!");
}
}catch(Exception e) {
e.printStackTrace();
}
}
}
Run: DirExist
/opt/module/jdk1.8.0_171/bin/java ...
newDir
2023-10-15 08:47:06,149 WARN [main] util.NativeCodeLoader (NativeCodeLoader.java:<clinit>(62)) - Unable
Directory Exists!
Process finished with exit code 0
Version Control Run TODO Problems Terminal Build Dependencies Event Log


hdfs_pro - DirExist.java
File Edit View Navigate Code Refactor Build Run Tools VCS Window Help
hdfs_pro src main java DirExist main
Project
hdfs_pro /headless/Desktop/
.idea
src
main
java
CreateDir
DirExist
ListFiles
resources
test
target
pom.xml
External Libraries
Scratches and Consoles
pom.xml (hdfs_pro)
CreateDir.java
DirExist.java
ListFiles.java
try {
Scanner sc = new Scanner(System.in);
String dirName = '/'+sc.next();
FileSystem fs = FileSystem.get(new URI(str: "hdfs://master:9000"
if(fs.exists(new Path(dirName))) {
System.out.println("Directory Exists!");
} else {
System.out.println("Directory not Exists!");
}
}catch(Exception e) {
e.printStackTrace();
}
}
}
Run: DirExist
/opt/module/jdk1.8.0_171/bin/java ...
abc
2023-10-15 08:47:44,289 WARN [main] util.NativeCodeLoader (NativeCodeLoader.java:<clinit>(62)) - Unable
Directory not Exists!
Process finished with exit code 0
Version Control Run TODO Problems Terminal Build Dependencies Event Log

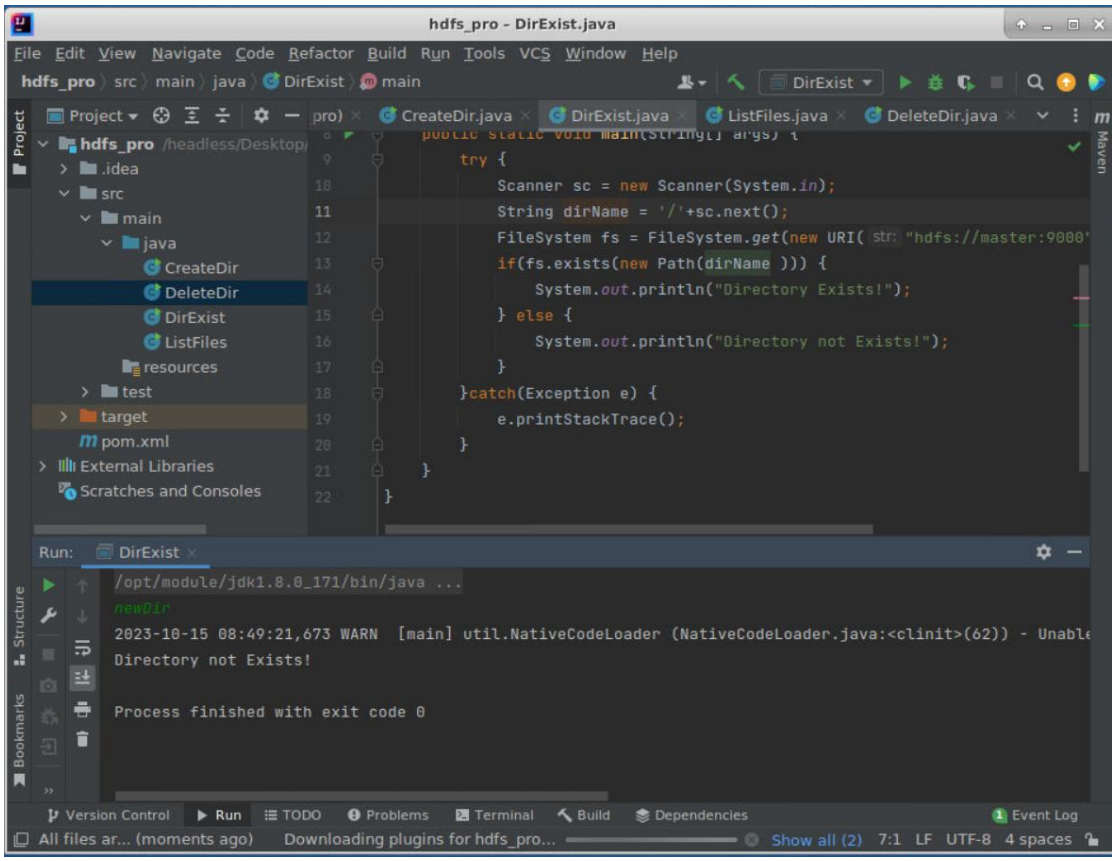
查看目录下文件



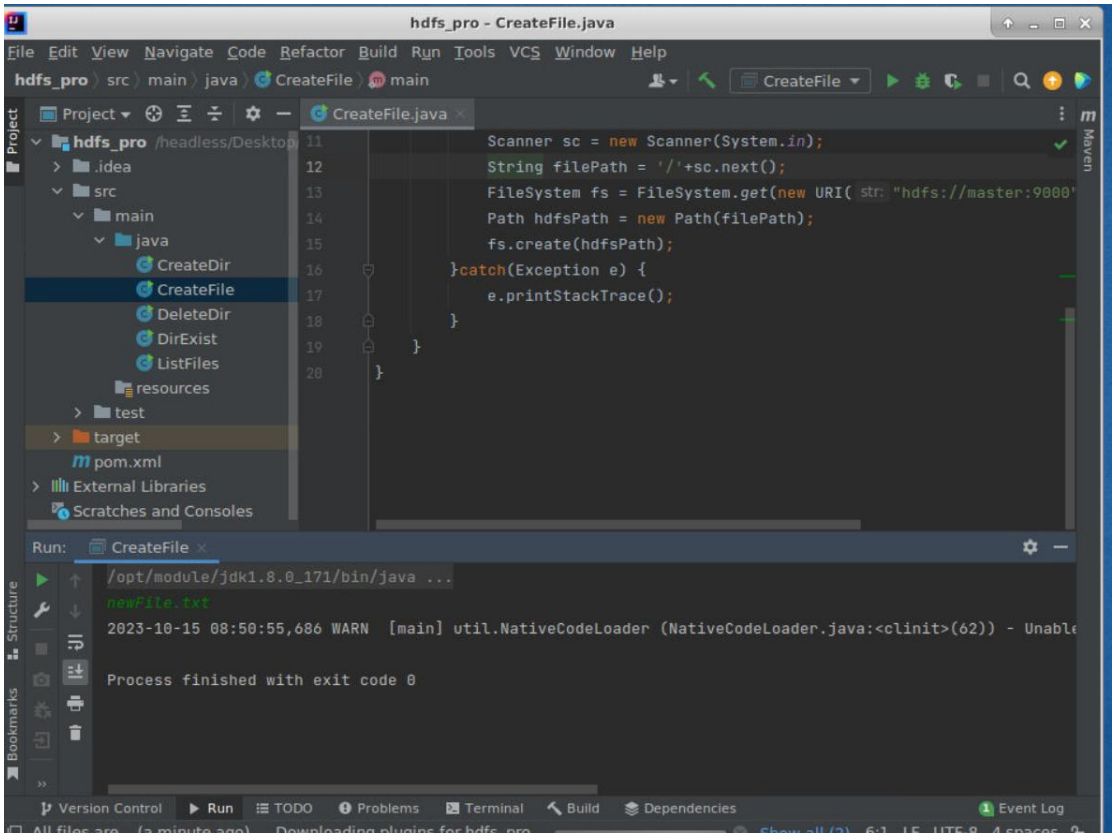
删除目录



成功删除，目录已不存在



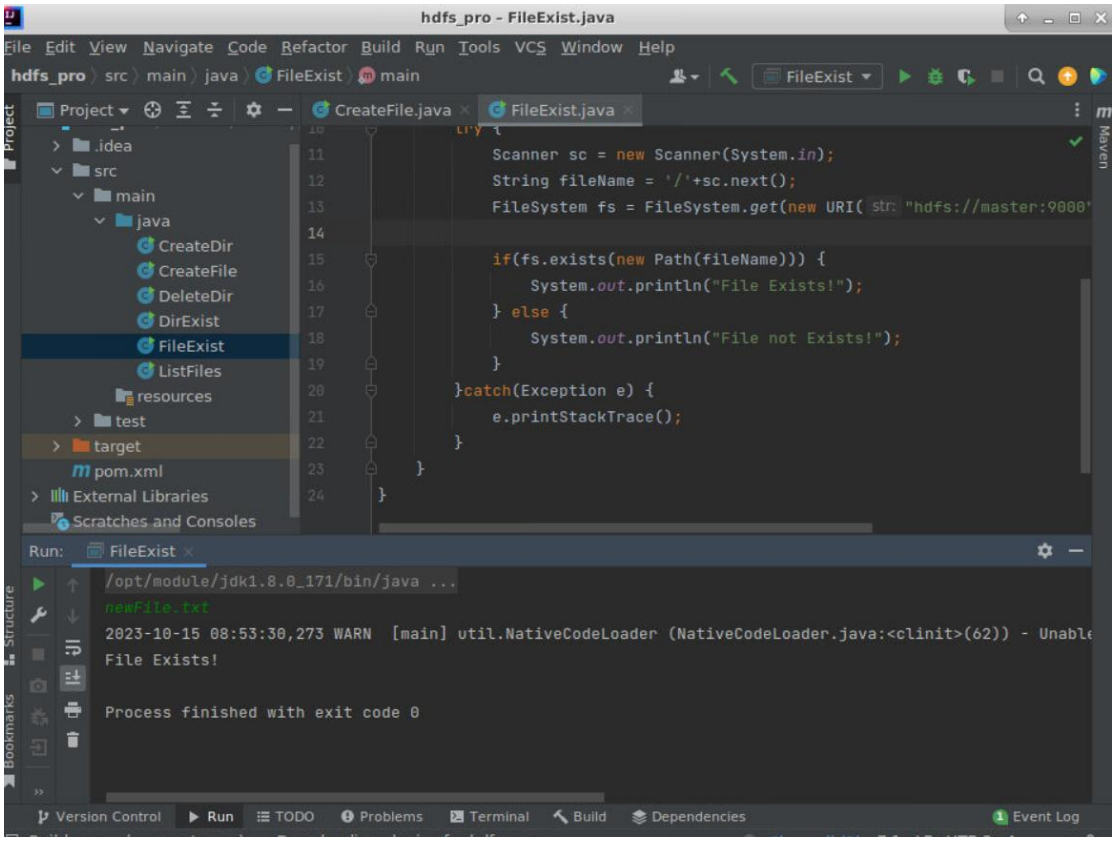
新建文件



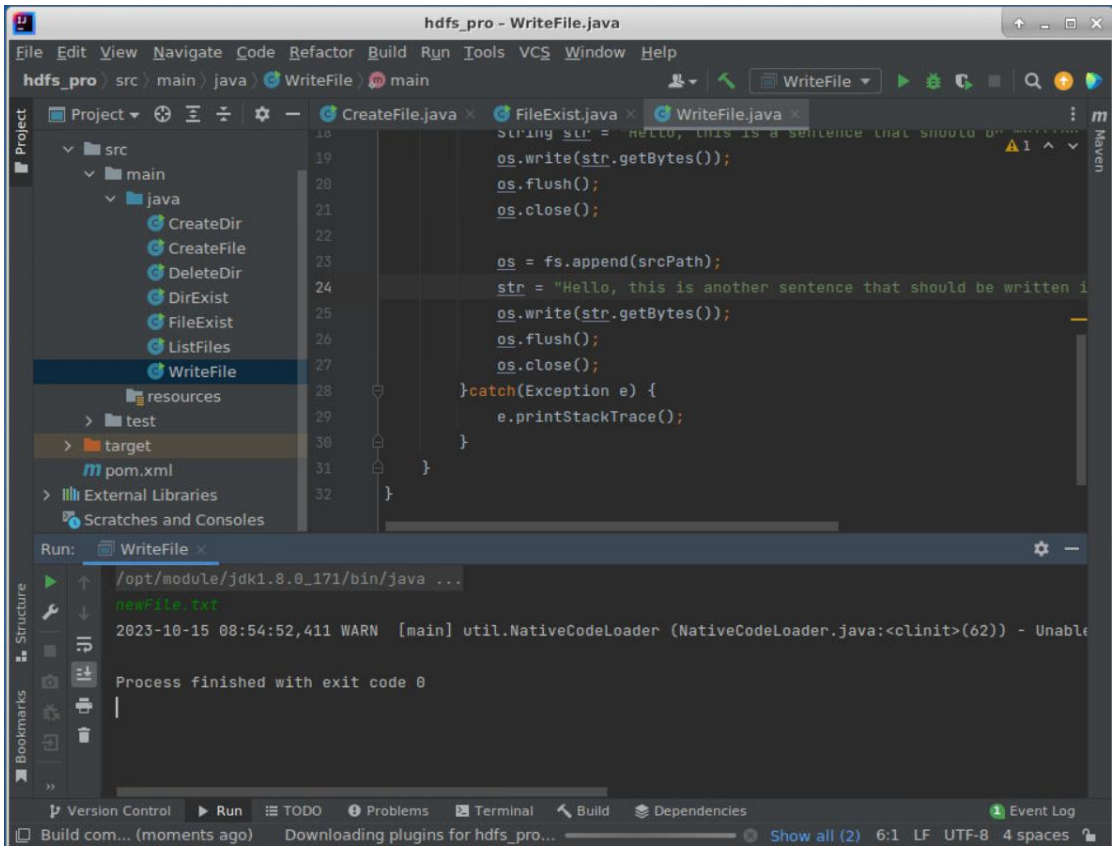
成功新建

```
root@master:~# hadoop fs -ls /
Found 3 items
drwxr-xr-x   - root supergroup          0 2023-10-11 08:42 /mydir
-rw-r--r--   3 root supergroup          0 2023-10-15 08:50 /newFile.txt
drwxrwx---   - root supergroup          0 2023-10-11 08:13 /tmp
root@master:~#
```

检测文件是否存在

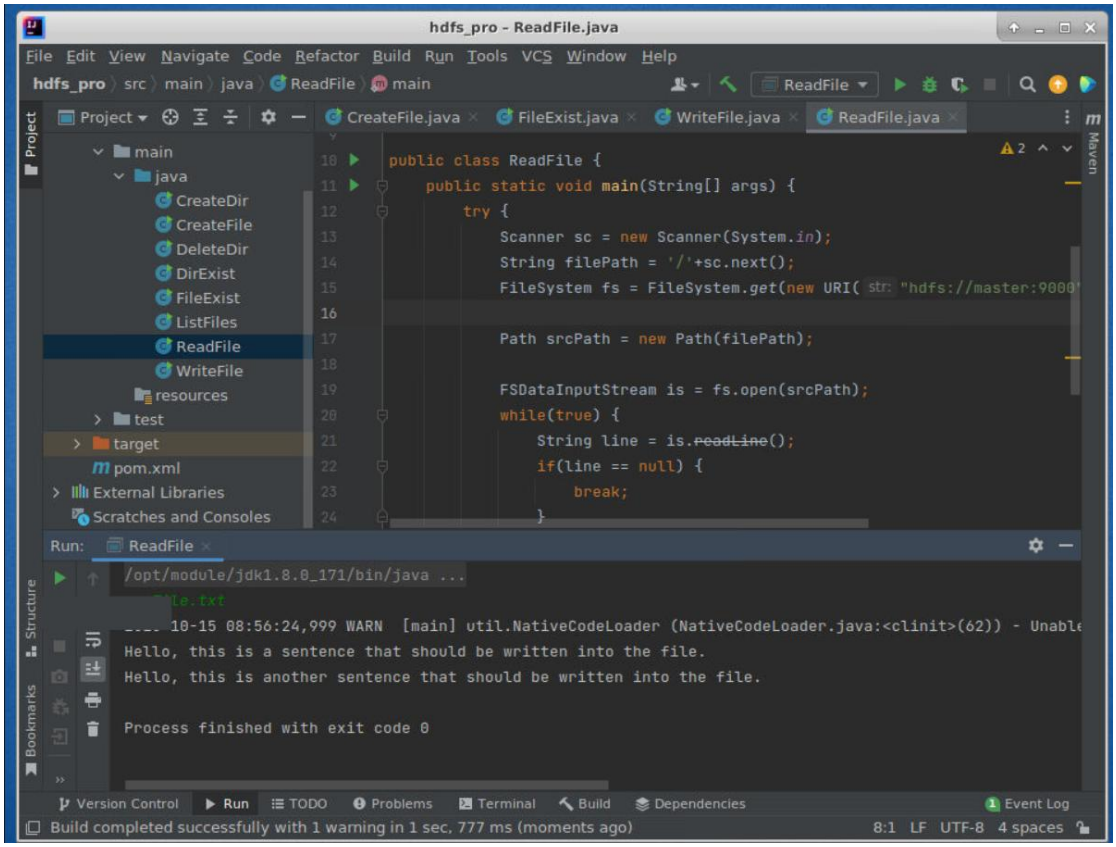


文件写

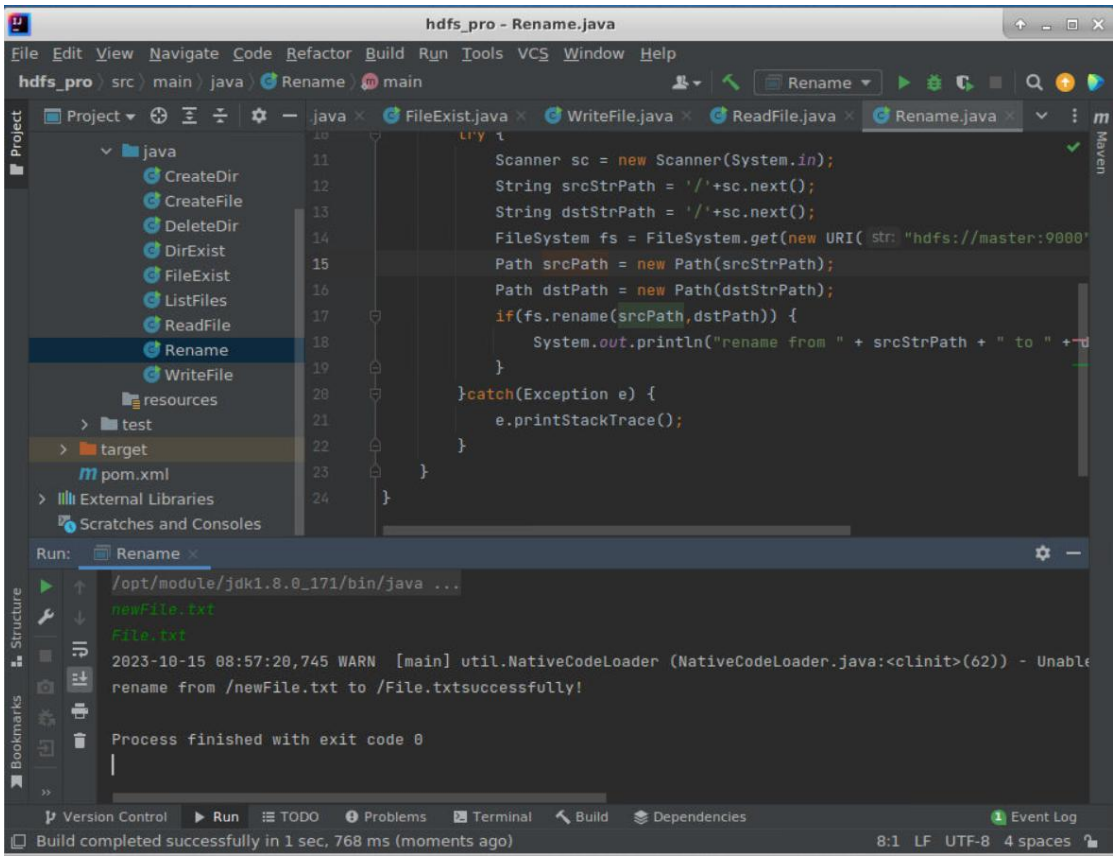



```
root@master:~# hadoop fs -cat /newFile.txt
Hello, this is a sentence that should be written into the file.
Hello, this is another sentence that should be written into the file.
root@master:~#
```

文件读

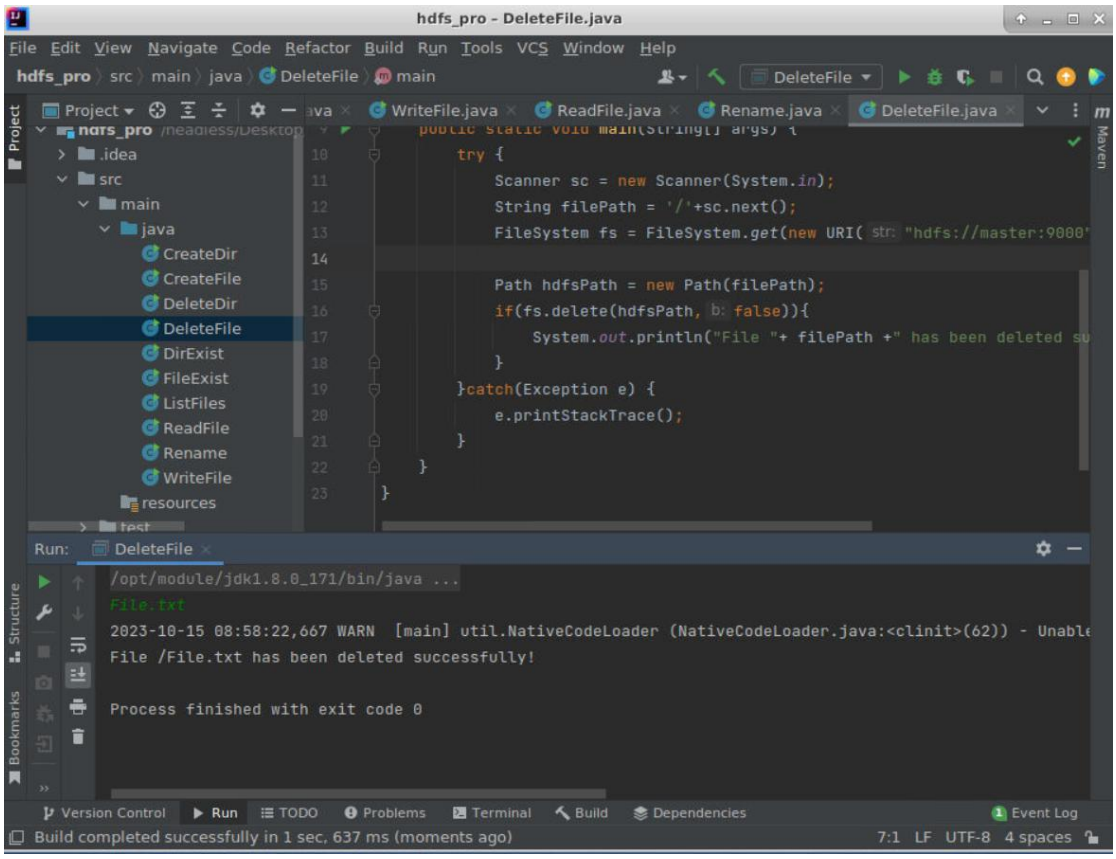

hdfs_pro - ReadFile.java
File Edit View Navigate Code Refactor Build Run Tools VCS Window Help
hdfs_pro src main java ReadFile main
CreateFile.java FileExist.java WriteFile.java ReadFile.java
main
java
CreateDir
CreateFile
DeleteDir
DirExist
FileExist
ListFiles
ReadFile
WriteFile
resources
test
target
pom.xml
External Libraries
Scratches and Consoles
public class ReadFile {
public static void main(String[] args) {
try {
Scanner sc = new Scanner(System.in);
String filePath = '/'+sc.next();
FileSystem fs = FileSystem.get(new URI(str: "hdfs://master:9000"
Path srcPath = new Path(filePath);
FSDataInputStream is = fs.open(srcPath);
while(true) {
String line = is.readLine();
if(line == null) {
break;
}
Run: ReadFile
/opt/module/jdk1.8.0_171/bin/java ...
10-15 08:56:24,999 WARN [main] util.NativeCodeLoader (NativeCodeLoader.java:<clinit>(62)) - Unable
Hello, this is a sentence that should be written into the file.
Hello, this is another sentence that should be written into the file.
Process finished with exit code 0
Version Control Run TODO Problems Terminal Build Dependencies Event Log
Build completed successfully with 1 warning in 1 sec, 777 ms (moments ago)
8:1 LF UTF-8 4 spaces

文件重命名


hdfs_pro - Rename.java
File Edit View Navigate Code Refactor Build Run Tools VCS Window Help
hdfs_pro src main java Rename main
java FileExist.java WriteFile.java ReadFile.java Rename.java
java
CreateDir
CreateFile
DeleteDir
DirExist
FileExist
ListFiles
ReadFile
Rename
WriteFile
resources
test
target
pom.xml
External Libraries
Scratches and Consoles
Scanner sc = new Scanner(System.in);
String srcStrPath = '/'+sc.next();
String dstStrPath = '/'+sc.next();
FileSystem fs = FileSystem.get(new URI(str: "hdfs://master:9000"
Path srcPath = new Path(srcStrPath);
Path dstPath = new Path(dstStrPath);
if(fs.rename(srcPath,dstPath)) {
System.out.println("rename from " + srcStrPath + " to " +
}
}catch(Exception e) {
e.printStackTrace();
}
}
}
Run: Rename
/opt/module/jdk1.8.0_171/bin/java ...
newFile.txt
File.txt
2023-10-15 08:57:20,745 WARN [main] util.NativeCodeLoader (NativeCodeLoader.java:<clinit>(62)) - Unable
rename from /newFile.txt to /File.txtsuccessfully!
Process finished with exit code 0
Version Control Run TODO Problems Terminal Build Dependencies Event Log
Build completed successfully in 1 sec, 768 ms (moments ago)
8:1 LF UTF-8 4 spaces

```
root@master:~# hadoop fs -ls /
Found 3 items
-rw-r--r--   1 root supergroup        134 2023-10-15 08:54 /File.txt
drwxr-xr-x   - root supergroup          0 2023-10-11 08:42 /mydir
drwxrwx---   - root supergroup          0 2023-10-11 08:13 /tmp
root@master:~#
```

删除文件




```
root@master:~# hadoop fs -ls /
Found 2 items
drwxr-xr-x   - root supergroup          0 2023-10-11 08:42 /mydir
drwxrwx---   - root supergroup          0 2023-10-11 08:13 /tmp
root@master:~#
```

实验作业

```
2023-10-24 07:53:51,493 WARN  [main] util.NativeCodeLoader (NativeCodeLoader.java:<clinit>(62)) - Unable
Hshell -list /
FileStatus{path=hdfs://master:9000/; isDirectory=true; modification_time=1697615888079; access_time=0; ow
Hshell -rm -r /mydir
Directory /mydir has been deleted successfully!
```

import org.apache.hadoop.conf.Configuration;

import org.apache.hadoop.fs.FSDataOutputStream;

```
import org.apache.hadoop.fs.FileStatus;

import org.apache.hadoop.fs.FileSystem;

import org.apache.hadoop.fs.Path;


import java.io.IOException;

import java.net.URI;

import java.util.Scanner;


public class HShell {

    public static void main(String[] args) {

        try {

            Scanner sc = new Scanner(System.in);

            FileSystem fs = FileSystem.get(new URI("hdfs://master:9000"), new Con
figuration());

            while(true){

                String cmd[] = sc.nextLine().split("\\s+");

                if(cmd[0].equals("Hshell")){

                    if(cmd[1].equals("-cp")){

                        if(cmd[2].equals("-r")){

                            fs.copyFromLocalFile(false,true,new Path(cmd[3]),new
Path(cmd[4]));

                            System.out.println("copy directory from "+cmd[3]+"to "+
cmd[4]);

                        } else {

                            fs.copyFromLocalFile(new Path(cmd[2]),new Path(cmd
[3]));
```

```
                    System.out.println("copy file from "+cmd[2]+"to "+cmd
[3]);
                }
            } else if (cmd[1].equals("-rm")){
                if(cmd[2].equals("-r")){
                    if(fs.delete(new Path(cmd[3]),true)){
                        System.out.println("Directory "+ cmd[3] +" has bee
n deleted successfully!");
                    }
                } else {
                    if(fs.delete(new Path(cmd[2]),false)){
                        System.out.println("File "+ cmd[2] +" has been del
eted successfully!");
                    }
                }
            } else if (cmd[1].equals("-list")){
                FileStatus filestatus =fs.getFileStatus(new Path(cmd[2]));
                System.out.println(filestatus.toString());
            } else if (cmd[1].equals("-mv")){
                fs.moveFromLocalFile(new Path(cmd[2]),new Path(cmd[3]));
                System.out.println("move file from "+cmd[2]+"to "+cmd[3]);
            } else if (cmd[1].equals("-find")){
                fs.setWorkingDirectory(new Path(cmd[3]));
                if(!fs.exists(new Path(cmd[2]))) {
                    System.out.println(cmd[2]+" is not found");
```

```
                    } else {
                        System.out.println(fs.getFileStatus(new Path(cmd[2])).ge
tPath());
                    }
                } else {
                    System.out.println("Error Input!\n");
                }
            } else if (cmd[0].equals("Exit")){
                System.out.println("Exit!\n");
                break;
            } else {
                System.out.println("Error Input!\n");
            }


        }
    }catch(Exception e) {
        e.printStackTrace();
    }
  }
}
```

五、实验总结

通过本次实验，我初步了解到了如何在 hdfs 上进行基础的文件、目录操作；学习到了如何借助 Hadoop API，编写 java 程序实现对 hdfs 的操作，并实现了一些简单的文件、目录操作。同时，也认识到 hdfs 有许多接口，十分便利，引起了我学习的兴趣。